



Tutorial S3: Normal Distribution

Section A (Basic Questions)

1 Given that $X \sim N(1.5, 2.5)$, evaluate

- (a) $P(X < 0.7)$, (b) $P(X \geq 3.7)$, (c) $P(1.1 < X < 2.3)$,
 (d) $P(|X| \leq 0.5)$, (e) $P(|X| > 0.5)$, (f) $P(|X - 1.5| < \sqrt{2.5})$.

[(a) 0.306, (b) 0.0821, (c) 0.293, (d) 0.161, (e) 0.839, (f) 0.683]

Solution:

(a)

NORMAL FLOAT AUTO REAL RADIAN MP
`normalcdf`
 lower: -E99
 upper: 0.7
 μ : 1.5
 σ : $\sqrt{2.5}$
 Paste

Note that for this last entry, we need to key in the standard deviation and not the variance!

Use $\text{normalcdf}(-E99, 0.7, 1.5, \sqrt{2.5})$

(b)

Use $\text{normalcdf}(3.7, E99, 1.5, \sqrt{2.5})$

(c)

Use $\text{normalcdf}(1.1, 2.3, 1.5, \sqrt{2.5})$

(d)

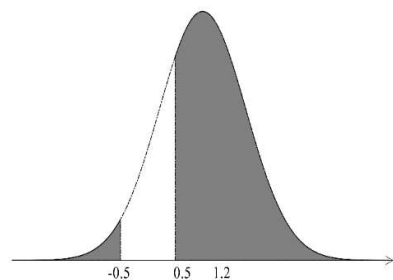
$P(|X| \leq 0.5) = P(-0.5 \leq X \leq 0.5)$

(e)

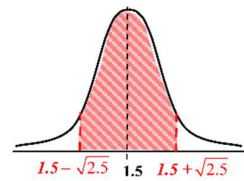
$P(|X| > 0.5) = 1 - P(-0.5 \leq X \leq 0.5)$
 $= 0.839$ (3 s.f.)

OR:

$P(|X| > 0.5) = P(X < -0.5) + P(X > 0.5)$



(f)
$$\begin{aligned} P(|X - 1.5| < \sqrt{2.5}) &= P(-\sqrt{2.5} < X - 1.5 < \sqrt{2.5}) \\ &= P(1.5 - \sqrt{2.5} < X < 1.5 + \sqrt{2.5}) \\ &= 0.683 \text{ (3 s.f.)} \end{aligned}$$



Note that since X has mean 1.5 and standard deviation $\sqrt{2.5}$, $P(|X - 1.5| < \sqrt{2.5})$ is the probability that X is within 1 standard deviation from its mean, so $P(|X - 1.5| < \sqrt{2.5}) = P(|Z| < 1)$.

Or using standardisation, we have:

$$\begin{aligned} P(|X - 1.5| < \sqrt{2.5}) &= P\left(\frac{|X - 1.5|}{\sqrt{2.5}} < \frac{\sqrt{2.5}}{\sqrt{2.5}}\right) \\ &= P\left(\left|\frac{X - 1.5}{\sqrt{2.5}}\right| < 1\right) \\ &= P(|Z| < 1) \end{aligned}$$

2 Given that $Y \sim N(22, 5)$ and $Z \sim N(0, 1)$, find the value(s) of k such that

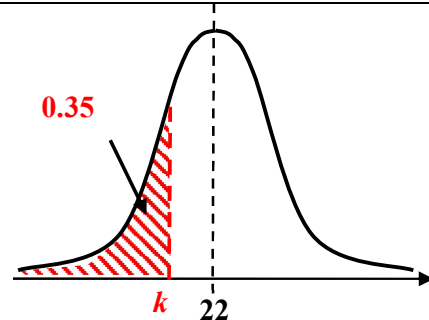
- (a) $P(Y < k) = 0.35$,
- (b) $P(Y > k) = 0.8$,
- (c) $P(k < Y < 25) = 0.46$,
- (d) $P(|Z| > k) = 0.27$,
- (e) $P(22 - k < Y < 22 + k) = 0.67$,
- (f) $P(Y < k) > 0.3$.

[(a) 21.1, (b) 20.1, (c) 21.7, (d) 1.10, (e) 2.18, (f) $k > 20.8$]

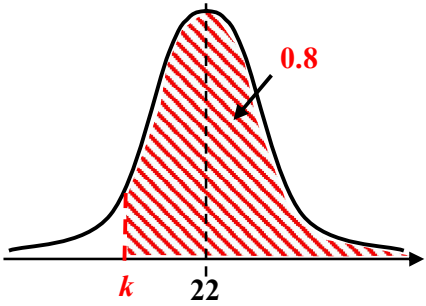
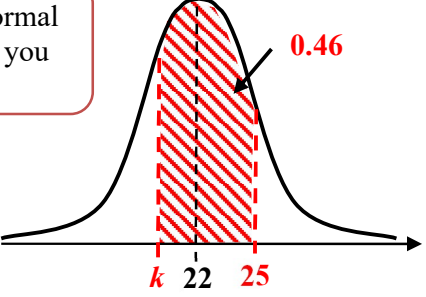
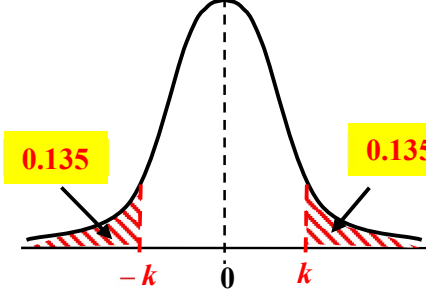
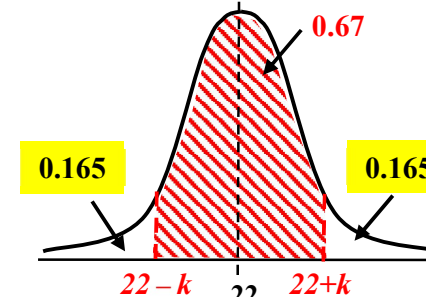
Solution:

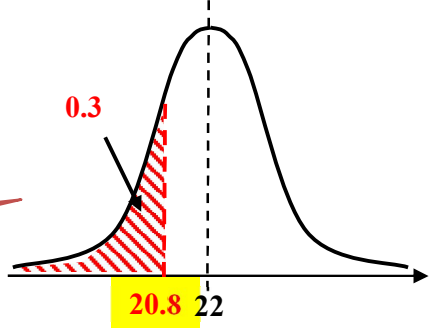
(a) $P(Y < k) = 0.35$

From G.C, $k = 21.1$ (3 s.f)



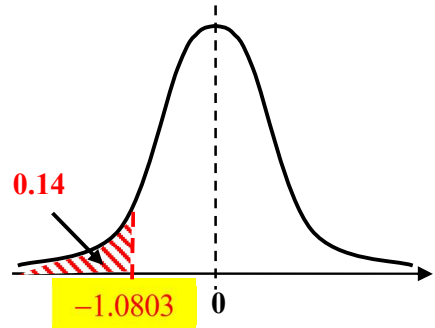
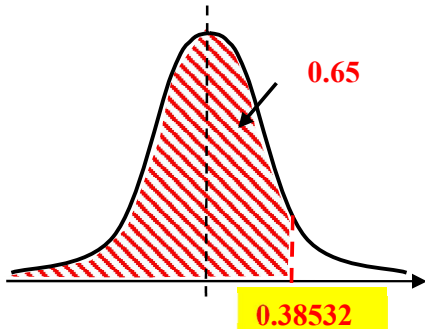
$$k = \text{invNorm}(0.35, 22, \sqrt{5}, \text{LEFT})$$

(b)	$P(Y > k) = 0.8$ From G.C, $k = 20.1$ (3 s.f)	 $k = \text{invNorm}(0.8, 22, \sqrt{5}, \text{RIGHT})$
(c)	$P(k < Y < 25) = 0.46$ From the diagram, $P(Y < 25) - P(Y < k) = 0.46$ $P(Y < k) = P(Y < 25) - 0.46$ $P(Y < k) = 0.45014$ From G.C, $k = 21.7$ (3 s.f)	Sketch the normal curve to help you visualise!  $k = \text{invNorm}(0.4501, 22, \sqrt{5}, \text{LEFT})$
(d)	$P(Z > k) = 0.27$ From the diagram, $P(Z \leq k) = 0.73$ $P(-k \leq Z \leq k) = 0.73$ From G.C, $k = 1.10$ (3 s.f) Use $\text{invNorm}(\dots \text{CENTER})$ since the lower bound $-k$ and upper bound k are symmetric about $E(Z) = 0$.	 $\text{invNorm}(0.73, 0, 1, \text{CENTER})$
(e)	$P(22 - k < Y < 22 + k) = 0.67$ From G.C, $22 - k = 19.822$ and $22 + k = 24.178$ $\therefore k = 2.18$ (3 s.f) Use $\text{invNorm}(\dots \text{CENTER})$ since the lower bound $22 - k$ and upper bound $22 + k$ are symmetric about $E(Y) = 22$.	 $\text{invNorm}(0.67, 22, \sqrt{5}, \text{CENTER})$

(f) $P(Y < k) > 0.3$ From G.C, $P(Y < 20.8) = 0.3$ Referring to the diagram, $k > 20.8$ (3 s.f) For such questions, draw the normal curve to help you visualize!	 $\text{invNorm}(0.3, 22, \sqrt{5}, \text{LEFT}) = 20.8$
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- 3 The random variable X follows a normal distribution with mean μ and standard deviation σ . Given that $P(X < 20) = 0.14$ and that $P(X < 50) = 0.65$, calculate the values of μ and σ .

[42.1, 20.5]

Solution: $X \sim N(\mu, \sigma^2)$ $P(X < 20) = 0.14$ $P\left(\frac{X - \mu}{\sigma} < \frac{20 - \mu}{\sigma}\right) = 0.14$ Recall that $Z = \frac{X - \mu}{\sigma}$ $P\left(Z < \frac{20 - \mu}{\sigma}\right) = 0.14$ $\frac{20 - \mu}{\sigma} = -1.0803$ from GC $1.0803\sigma - \mu = -20 \quad \dots(1)$ $P(X < 50) = 0.65$ $P\left(\frac{X - \mu}{\sigma} < \frac{50 - \mu}{\sigma}\right) = 0.65$ $P\left(Z < \frac{50 - \mu}{\sigma}\right) = 0.65$ $\frac{50 - \mu}{\sigma} = 0.38532$ from GC $0.38532\sigma + \mu = 50 \quad \dots(2)$	 $\text{invNorm}(0.14, 0, 1, \text{LEFT}) = -1.0803$  $\text{invNorm}(0.65, 0, 1, \text{LEFT}) = 0.38532$
Solving (1) and (2), $\mu = 42.1$ (3sf) and $\sigma = 20.5$ (3sf)	

- 4 The independent random variables X and Y are each normally distributed with means 6 and 8 respectively, and variances 9 and 16 respectively. Find
- (a) (i) $P(X < Y)$, (ii) $P(|X - Y| < 1.5)$.
- (b) X_1, X_2 and X_3 are three independent observations of X , and Y_1 and Y_2 are two independent observations of Y . Find
- (i) $P(X_1 + X_2 + X_3 < 15)$,
- (ii) $P(X_1 + X_2 > Y_1 + Y_2)$,
- (iii) $P(2X > Y_1 + Y_2)$.

[(a)(i) 0.655, (ii) 0.218, (b)(i) 0.282, (ii) 0.286, (iii) 0.314]

Solution:	
Given $X \sim N(6, 3^2)$ and $Y \sim N(8, 4^2)$	
(a) (i)	$P(X < Y) = P(X - Y < 0)$ $E(X - Y) = E(X) - E(Y) = 6 - 8 = -2$ $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) = 9 + 16 = 25$ $\therefore X - Y \sim N(-2, 25)$ $P(X - Y < 0) = 0.655 \quad (3 \text{ s.f.})$
	Rewrite the inequality so that all random variables are on the left. Note that we add the variances, never subtract.
(a) (ii)	$P(X - Y < 1.5) = P(-1.5 < X - Y < 1.5)$ $= 0.218 \quad (3 \text{ s.f.})$
	Write down the distribution before finding the probability.
(b) (i)	$E(X_1 + X_2 + X_3) = 3E(X) = 18$ $\text{Var}(X_1 + X_2 + X_3) = 3\text{Var}(X) = 27$ $\therefore X_1 + X_2 + X_3 \sim N(18, 27)$ $P(X_1 + X_2 + X_3 < 15) = 0.282 \quad (3 \text{ s.f.})$
(b) (ii)	$P(X_1 + X_2 > Y_1 + Y_2) = P(X_1 + X_2 - (Y_1 + Y_2) > 0)$ $E[X_1 + X_2 - (Y_1 + Y_2)] = 2E(X) - 2E(Y) = 12 - 16 = -4$ $\text{Var}[X_1 + X_2 - (Y_1 + Y_2)] = 2\text{Var}(X) + 2\text{Var}(Y) = 2(9) + 2(16) = 50$ $\therefore X_1 + X_2 - (Y_1 + Y_2) \sim N(-4, 50)$ $P(X_1 + X_2 - (Y_1 + Y_2) > 0) = 0.286 \quad (3 \text{ s.f.})$
(b) (iii)	$P(2X > Y_1 + Y_2) = P(2X - (Y_1 + Y_2) > 0)$ $E[2X - (Y_1 + Y_2)] = 2E(X) - 2E(Y) = 12 - 16 = -4$ $\text{Var}[2X - (Y_1 + Y_2)] = 2^2\text{Var}(X) + 2\text{Var}(Y) = 4(9) + 2(16) = 68$ $\therefore 2X - (Y_1 + Y_2) \sim N(-4, 68)$ $P(2X - (Y_1 + Y_2) > 0) = 0.314 \quad (3 \text{ s.f.})$